

3(a)	<table><tr><th rowspan="2">event</th><th colspan="4">stage</th><th rowspan="2"></th></tr><tr><th>G</th><th>LR</th><th>KC</th><th>OP</th></tr><tr><td>catabolism</td><td>✓</td><td>✓</td><td>✓</td><td>✗</td><td>;</td></tr><tr><td>coenzyme is reduced or oxidised</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>;</td></tr><tr><td>a coenzyme forms a covalent bond with a respiratory intermediate</td><td>✗</td><td>✓</td><td>✗</td><td>✗</td><td>;</td></tr><tr><td>carbon dioxide is released</td><td>✗</td><td>✓</td><td>✓</td><td>✗</td><td>;</td></tr></table>	event	stage					G	LR	KC	OP	catabolism	✓	✓	✓	✗	;	coenzyme is reduced or oxidised	✓	✓	✓	✓	;	a coenzyme forms a covalent bond with a respiratory intermediate	✗	✓	✗	✗	;	carbon dioxide is released	✗	✓	✓	✗	;	4
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3(b)(i)	any four from: 1 (RQ =) ratio of carbon dioxide (exhaled) to oxygen (inhaled) ; 2 air flow meter measures oxygen inhaled ; 3 carbon dioxide sensor measures concentration of CO₂ exhaled ; 4 carbohydrate RQ = 1 ; 5 lipid RQ = 0.7 ;	4																																		
3(b)(ii)	any three from: 1 lipid has higher energy (value) OR carbohydrate 15–17 kJ g⁻¹ and lipid 37–40 kJ g⁻¹ ; 2 more, hydrogen (atoms) / C-H bonds, in lipid ; 3 more reduced, NAD / FAD ; 4 greater proton gradient / more protons in intermembrane space / more protons pass through ATP synth(et)ase ;	3																																		